Worksheet 5: Terrestrial Planets

Covers Lectures 9 and 10: equilibrium temperature, the runaway greenhouse, deuterium and water loss, Jeans escape, Mercury and Mars

Planetary Systems (WBAS002-05) • Kapteyn Astronomical Institute, University of Groningen

This worksheet is ungraded practice, with the same shape and level as the final exam. Plan up to 2 hours for a serious pass. Work each problem through on paper before you open the solutions, and show your algebra: the exam asks for the steps, not only the number. Some parts state an intermediate result ("show that …"); use it to check your work, and to continue if a step resists. Carry at least four significant figures through the intermediate steps (the worked solutions print five) and round only at the end. All parts are analytical; no computer is needed.

Constants and data

$\sigma = 5.670 \times 10^{-8}\ \mathrm{W\,m^{-2}\,K^{-4}}$, $k_B = 1.381 \times 10^{-23}\ \mathrm{J\,K^{-1}}$, $G = 6.674 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$, $m_u = 1.661 \times 10^{-27}$ kg, $S_0 = 1361\ \mathrm{W\,m^{-2}}$ (at 1 AU)

Water and grey atmosphere: latent heat $L = 2.5 \times 10^6\ \mathrm{J\,kg^{-1}}$, vapour gas constant $R_v = 461\ \mathrm{J\,kg^{-1}\,K^{-1}}$, reference $(p_0, T_0) = (611\ \mathrm{Pa}, 273\ \mathrm{K})$, grey opacity $\kappa = 5 \times 10^{-2}\ \mathrm{m^2\,kg^{-1}}$, $g = 10\ \mathrm{m\,s^{-2}}$

Earth: $A = 0.30$, $T_{surf} = 288$ K, $a = 1.00$ AU. Venus: $A = 0.77$, $a = 0.723$ AU, $S = 1.91\,S_0$, $T_{surf} = 737$ K

Mars: $M = 6.4 \times 10^{23}$ kg, exobase radius $r_{exo} = 3.6 \times 10^6$ m, exobase temperature $T_{exo} = 270$ K, rotation period 24.62 h

Mercury: $R = 2440$ km, spin period $P_{rot} = 58.65$ d, orbital period $P_{orb} = 87.97$ d, radius change $\Delta R \approx 7$ km. Phobos: orbital radius $a = 9376$ km

Relative molecular masses (in units of m_u): H = 1, H_2 = 2, CO_2 = 44

Atmospheric and escape data as in the Lecture 9 and 10 notes. A global equivalent layer (GEL) is the depth of a uniform global water layer holding a given mass; take Earth's ocean as ≈ 2700 m GEL and Venus's current atmospheric water as ≈ 0.02 m GEL.

Problem 1 Equilibrium temperature and the greenhouse effect

The equilibrium temperature is the first number to compute for any planet: it is what the surface would reach with no atmosphere, set by the balance of absorbed sunlight against emitted infrared. The gap between it and the true surface temperature measures the greenhouse effect.

(a) The energy balance for a rapidly rotating planet of Bond albedo A at insolation S is $\pi R^2 S(1-A) = 4\pi R^2 \sigma T_{eq}^4$, so $T_{eq} = [S(1-A)/4\sigma]^{1/4}$. Compute Earth's equilibrium temperature.

(b) A single infrared-opaque layer in radiative equilibrium re-radiates downward as well as upward, so the surface energy balance becomes $\sigma T_s^4 = 2\sigma T_{eq}^4$. Show that $T_s = 2^{1/4}\,T_{eq}$, evaluate it for Earth, and compare with the observed surface temperature of 288 K.

(c) Compute Venus's equilibrium temperature from its own albedo and insolation, compare it with Earth's, and state the greenhouse warming $\Delta T = T_{\text{surf}} - T_{\text{eq}}$ for each planet.

(d) A fellow student writes: "Venus is hotter than Earth mainly because it is closer to the Sun." Judge this claim, using your numbers from parts (a) and (c).

Problem 2 The runaway greenhouse

A planet stays habitable only while its atmosphere can radiate away the sunlight it absorbs. A moist atmosphere has a ceiling on how much infrared it can emit, and crossing it triggers a runaway greenhouse. This problem locates that ceiling and applies it to early Venus.

(a) Consider a young terrestrial planet with liquid water and Earth-like water clouds, so $A = 0.30$. Compute the absorbed solar flux per unit surface area, $F_{\text{abs}} = S(1 - A)/4$, for a planet at Earth's orbit and for one at Venus's orbit.

(b) In a steam atmosphere the infrared photosphere forms where the overlying water-vapour column reaches optical depth of order unity, at pressure $p_{\text{phot}} = g/\kappa$. Compute p_{phot}, then use the integrated Clausius-Clapeyron relation

$$\ln \frac{p}{p_0} = -\frac{L}{R_v}\left(\frac{1}{T} - \frac{1}{T_0}\right)$$

to find the temperature at which saturated water vapour reaches this pressure.

(c) The outgoing longwave radiation of the steam atmosphere cannot exceed the black-body emission of its photosphere, $F_{\text{OLR}}^{\text{max}} = \sigma T_{\text{phot}}^4$. Compute this limit, compare it with the absorbed fluxes from part (a), and state which planet enters a runaway greenhouse.

(d) Explain in words why the outgoing infrared flux of a steam atmosphere saturates at an upper limit, and why this defines the inner edge of the habitable zone.

(e) Venus's present albedo is $A = 0.77$, high enough that at its actual albedo it absorbs less than the limit you found in part (c). Explain why this does not contradict the conclusion that Venus underwent a runaway greenhouse.

Problem 3 Deuterium and Venus's lost water

Venus's atmosphere is enriched in deuterium relative to ordinary hydrogen by about a factor of 150 compared with Earth's oceans. This fingerprint of preferential hydrogen escape lets us estimate how much water Venus began with. The isotope ratio in the remaining water follows the Rayleigh distillation law $R/R_0 = f^{\alpha-1}$, where f is the fraction of the original hydrogen inventory that remains, R/R_0 is the deuterium enrichment, and α is the fractionation factor (the ratio of the deuterium to the hydrogen escape efficiency). Diffusion-limited escape gives $\alpha = 0.5$. Take the measured enrichment $R/R_0 = 150$.

(a) Solve the Rayleigh law for the fraction of the original water that remains.

(b) Convert to an initial water inventory in global-equivalent-layer depth, and compare it with Earth's ocean.

(c) Explain physically why the water left behind is enriched in deuterium, and what the enrichment would approach if essentially all the hydrogen escaped.

(d) A fellow student writes: "The factor-of-150 enrichment proves Venus once had a full ocean like Earth's." Judge this claim, using the sensitivity of the estimate to the escape model.

Problem 4 Thermal escape from Mars

Whether a planet keeps its atmosphere is decided at the exobase, the level where a molecule makes its last collision and the fastest ones escape. The Jeans parameter compares gravity's grip on a molecule to its thermal energy, and it sorts which gases a planet can hold.

(a) Compute the escape speed at Mars's exobase.

(b) The Jeans escape parameter for a molecule of mass m is $\lambda = GMm/(k_B T_{exo}\, r_{exo})$, the ratio of gravitational to thermal energy at the exobase. Compute λ for atomic hydrogen, molecular hydrogen, and carbon dioxide.

(c) The Jeans escape flux carries a factor $(1+\lambda)\,e^{-\lambda}$ and a thermal-speed factor $\propto m^{-1/2}$. Form the ratio of the escape fluxes of atomic and molecular hydrogen, and state whether CO_2 can escape thermally at all.

(d) A fellow student writes: "If Mars had kept a stronger magnetic field, it would have retained its CO_2 atmosphere." Judge this claim.

Problem 5 Mercury and Mars: two ways to end

Mercury and Mars sit at the inner and outer edges of the terrestrial planets. They diverged completely. This problem collects four independent measurements, each marking one step in that divergence.

(a) Mercury rotates on its axis every 58.65 days and orbits the Sun every 87.97 days. The length of a solar day, from noon to noon, satisfies $1/T_{solar} = 1/P_{rot} - 1/P_{orb}$. Compute T_{solar} and express it in Mercury years.

(b) As Mercury's interior cooled, its radius shrank by $\Delta R \approx 7$ km. Estimate the fractional decrease in surface area, and connect it to the lobate scarps seen across the surface.

(c) Phobos orbits Mars at a radius of 9376 km. Compute its orbital period from Kepler's third law, compare it with Mars's rotation period of 24.62 h, and state what this predicts for the moon's future.

(d) Mercury still has a magnetic field and a large iron core; Mars lost its field early and has a smaller core fraction. In two or three sentences, connect planet size to this divergence, using the ideas of cooling rate and atmospheric escape from this worksheet.